

POINT ESTIMATION

15th August 2026

HOW DO WE SOLVE THIS PROBLEM? TAKE YOUR BEST GUESS

In statistical problems, we often model uncertainty via probability distributions. However, on paper, we often assume that the parameters governing a distribution are known, whilst in practice this is rarely the case. Instead, we are typically given a set of observations and must infer the unknown parameters most consistent with the data available. Parameter estimation is simply the process of estimating unknown model parameter values derived from a model and experimental data.

For example, suppose that the return of an asset is believed to follow a Normal/Gaussian distribution, that is $X \sim N(\mu, \sigma^2)$. Whilst the form of the distribution may be known, the values of the parameters, μ and σ^2 , are generally unknown. Given a set of observed returns, $x_1, x_2, \dots, x_n$, we want to estimate these parameters using the available data. In general, if a probability distribution depends upon an unknown parameter, θ , then parameter estimation aims to construct an estimator, denoted by $\hat{\theta}$, that approximates the true value.

In many cases, there is inherent uncertainty in the best estimate of parameter values, this nature of uncertainty can be critical for ensuring that aspects of a model are not being overlooked, that is mathematically, that the data used provides a sufficient level of confidence. Whereas probability theory studies the behaviour of data generated from a known model, statistics, and uncertainty estimated from the current model, can be used to make subsequent improvements the model itself from observed data. By estimating parameters and quantifying the uncertainty associated with those estimates, we can improve models through design optimisation, ensuring an optimal solution, assess their reliability, and make more informed decisions. Do note that programming software, such as R or Python, can be used for model-based parameter estimation, and when implemented can be incredibly useful in both industry and research.

There are two main types of parameter estimation:

1. Point Estimation
2. Interval Estimation

POINT ESTIMATION

Point estimation involves the use of sample data to arrive at a single value, known as a statistic, which is there to serve as a best guess for an unknown population parameter. Point estimation can be contrasted with interval estimation, which is the use of sample data to calculate an interval of possible, or probable, values of an unknown population parameter. When the expectation of the estimator is the true value of θ , it is called an unbiased estimator of θ , mathematically $\mathbb{E}[\hat{\theta}] = \theta$. When the estimate converges in some sense to the true value of θ , we say that it is a consistent estimator of θ .

There are multiple methods to obtain a point estimate, the most common and famous are:

1. Maximum Likelihood Estimator (MLE)
2. Method of Moments Estimator (MME)
3. Minimum Variance Unbiased Estimator (MVUE)

In the next article, we will discuss the **Maximum Likelihood Estimator**.

In Article No.16, we discussed what a moment is in mathematical terms. Recall that the k^{th} raw moment of a probability distribution, F , is defined as $\mu_k = \mathbb{E}[X^k]$. The **Method of Moments Estimator**, MME, simply states that the k^{th} sample moment is defined as $\hat{\mu}_k = \frac{1}{n} \sum_{i=1}^n X_i^k$, and thus, $\hat{\mu}_k$ can be used as an estimate of μ_k .

Here is an example of the MME for a non-standard distribution. Suppose $X_1, X_2, \dots, X_n$ are n IID discrete R.Vs

with the PMF: $p(x) = \begin{cases} p, & x = 0 \\ 2p, & x = 1 \\ 1 - 3p, & x = 2 \end{cases}$. Thus, since $\mu_1 = \mathbb{E}[X] = 2 - 4p$, we have $\hat{p} = \frac{1}{4}(2 - \hat{\mu}_1) = \frac{1}{4}(2 - \bar{X})$.

Note that, for any sample $X_1, X_2, \dots, X_n$, with mean μ and variance σ^2 , $\bar{X}$ is an unbiased estimator for μ . However, $\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ is not an unbiased estimator for σ^2 , but, $S^2 := \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is an unbiased estimator for σ^2 .

Point estimates could be anything, for example, 0 is always a valid point estimate for any parameter; so is a million. Hence, we need to use some criteria by which an estimate could be considered better than

another. The MLE is unique in most situations but sometimes, it's not always obvious as to why it is any good. One possible criterion to consider an estimate to be good is unbiasedness, another is consistency. In general, unbiased estimates are considered as good estimates, but they are not unique either. For example, both sample mean and variance are unbiased estimates for the Poisson parameter. Thus, we need something more, one commonly used optimality criterion is to consider the unbiased estimate with the minimum variance, *i.e.*, the **Minimum Variance Unbiased Estimator**, MVUE, which is unique. We can utilise the Crammer-Rao inequality to find that under smoothness assumptions on a probability function $f(x; \theta)$, then $Var(T) \geq \frac{1}{-\mathbb{E}\left[\frac{\partial^2 L(\theta)}{\partial \theta^2}\right]}$. This

bound is known as the Crammer-Rao lower bound, CRLB. If the an unbiased estimate attains this lower bound, then it is the unique MVUE.

The MLE for the Poisson distribution, which is shown in Article No.18, is $\hat{\lambda} = \bar{X}$ and it is unbiased for λ . Now, $-\mathbb{E}\left[\frac{\partial^2 L(\lambda)}{\partial \theta^2}\right] = -\mathbb{E}\left[-\frac{\sum_{i=1}^n X_i}{\lambda^2}\right] = \frac{n}{\lambda}$. Hence, the CRLB is $\frac{\lambda}{n}$, which is also the variance of $\bar{X}$. Thus, here the MLE is also the MVUE.

CONCLUSION

When exploring many statistical models, we have unknown parameters. By estimating parameters via point estimation and quantifying the uncertainty, we can optimise models and find a best estimate for the unknown parameters.

ALEXANDER FLANN